

RITHIK MARUDAPPA

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EXPERIENCE

Product Engineer - Tabulation & Reporting Engine

August 2025 – Present

Metaforms
Bangalore, India

- Sole engineer on the organization's Tabulation Engine — owned product, architecture, engineering, and client demos end-to-end, delivering a production-grade tabs system now in active pilot with enterprise clients.
- Built the core compute engine from scratch — a custom Python SDK over DuckDB exposing tabulation operations as agent-configurable scripts, enabling agentic table creation from uploaded specs.
- Developed an AI-powered tab-spec parser and normalizer that extracts structural information from uploaded documents into clean inputs for automated table generation.
- Built an automated crosstab generation system benchmarked against industry tools (WinCross, Q/Displayr), supporting statistical significance testing, data weighting, confidence intervals, and complex tab-spec processing.
- Led RapidFlow (Studio), automating market-research operations end-to-end, a chat-driven system that spins up isolated agent-harness instances (Claude Code) on Cloudflare Sandbox over a pre-built survey boilerplate letting users generate, configure, and deploy full surveys through natural-language chat (Replit/Lovable for surveys).
- Architected the Data Processing Module — a multi-agentic system with task-parallelized LLM workflows for automated script generation (including SPSS), data parsing, and structured content output.
- Achieved 40% reduction in pipeline latency by optimizing core architecture with caching, lazy compute, task parallelization, and batch processing across large analytical datasets.

Associate Software Engineer

July 2024 - July 2025

SAP Labs (Analytics Cloud)
Bangalore, India

- Contributed to SAP Analytics Cloud's OEM Integration team, working on Query Builder and Orca Integration Services — data-intensive applications that present complex analytical results to non-technical business users.
- Implemented asynchronous batch processing enabling parallel analytics calls, reducing story page load times by ~18% on large datasets.
- Played a key role in the HANA Cloud migration of OIS, including full-cycle implementation of features like Time Filters from proof-of-concept to production deployment.
- Built a performance-tracking dashboard with D3.js charting and analytics, introducing the Collaboration Innovation Index (CII) for quantitative progress assessment across teams.

PROJECTS

Intelligent Ticket Assignment System Using RAG & Vector Embeddings

Python, Flask, Pinecone, OpenAI, Scikit-learn

- Leveraged OpenAI's text-embedding-ada-002 to embed support tickets into vector space, storing embeddings in Pinecone for semantic similarity search and automated assignment based on expertise and historical resolution data.
- Implemented a ranking pipeline with a lightweight ML model (Logistic Regression) to score candidates based on experience, current workload, and domain relevance — an LLM-powered content routing system.

SVEEP, Government Of India

District Collectorate Office, Hanumakonda

Hanumakonda, India
Mar 2024 - Apr 2024

- Led a team project that served 500+ unique users during Telangana state election in Hanumakonda.
- Designed and developed a WhatsApp bot capable of processing EPIC IDs and a captcha verification system using native WhatsApp API and endpoints by the Electoral Commission of India, providing polling booth information.
- Developed an interactive website using Firebase and React.js, empowering users to generate custom prompts and images for sharing, fostering voter participation and increasing user interaction. Website: voterspointhkn.in

EDUCATION

National Institute of Technology, Warangal

Warangal, India

2020 – 2024

B.Tech in Computer Science Engineering | CGPA: 8.10 / 10 | SAT Subject Test - 800/800 (Maths, Physics and Chemistry)

Class XII and X, Central Board of Secondary Education

Jeddah, Saudi Arabia

CBSE 12th : 97.5% | CBSE 10th : 94.5%

TECHNICAL SKILLS

AI & Agentic: Multi-agent orchestration, MCP, agent harnesses (Claude Code, Codex, OpenCode), Cursor, LLM pipelines, prompt & skill engineering, RAG, Pinecone, Scikit-learn

Data & Backend: DuckDB, PostgreSQL, HANA (columnar), SQLite, MongoDB, Firebase, FastAPI, Node.js, Express.js, Spring Boot

Frontend & Visualization: React.js, D3.js, data visualization

Cloud & DevOps: AWS (S3, Lambda, EC2), Cloudflare (Sandbox), Docker, CI/CD, PostHog, SAP BTP, Linux, Git